

Learning Objective Matrix

Digital STEM Content Delivery Platform

Project Details

Field	Details
Project Name	Digital STEM Content Delivery Platform
Client	ABC School
Document Type	Learning Objective Matrix
Prepared By	Business Analyst
Version	1.0
Date	00 000 000

1. Document Purpose

This Learning Objective Matrix maps each planned STEM and Coding content item to its class level, subject, topic, learning objective, content type, teacher usage and validation owner. It helps the Product Owner, Subject Teachers, Content Team, Developer Team and QA Tester maintain a shared understanding of why each content item is required and how it will support classroom teaching.

2. Document Inputs

- Approved Business Requirements Document
- Functional Requirements Document
- Business Rules Document
- Use Case Document
- Data Field List
- Role and Permission Matrix
- Content Requirement Brief
- Subject Teacher and SME input

3. Matrix Scope

The matrix covers the initial release learning content for teacher-led classroom delivery. It does not include student login, student progress tracking, student device usage, AR, VR headset-based learning or complex AI recommendation features.

4. Learning Objective Matrix

Content ID	Class Level	Subject / Topic	Learning Objective	Content Type	Validation Owner
LOM-01	Class 6 / 7	Chemistry: H ₂ O Molecule	Teacher should explain that a water molecule has two hydrogen atoms and one oxygen atom with a basic bond structure.	3D Video / Interactive 3D	Chemistry Subject Teacher / SME
LOM-02	Class 7 / 8	Physics: Electric Circuit	Teacher should explain basic circuit components and show the difference between open and closed circuits.	Animated Video / Interactive Web Content	Physics Subject Teacher
LOM-03	Class 8 / 9	Mathematics: 3D Geometry	Teacher should explain 3D shapes, faces, edges and vertices using clear visual examples.	Visual Explanation / 3D Animation	Mathematics Subject Teacher
LOM-04	Class 7 / 8	Computer Science: If-Else Logic	Teacher should explain decision-making logic using a simple condition, true path, false path and output.	Coding Simulation	Computer Science Subject Teacher / SME
LOM-05	Class 8 / 9	Computer Science: Loops in Programming	Teacher should explain repeated execution using simple loop examples and step-by-step output.	Coding Simulation	Computer Science Subject Teacher / SME

5. Detailed Learning Objective Mapping

Content ID	Key Concept	Teacher Usage	Expected Learner Understanding	Validation Check
LOM-01	Molecular structure of water	Teacher uses 3D view to rotate and label atoms.	Students understand two hydrogen atoms and one oxygen atom form H ₂ O.	Atom count, bond labels and molecule structure must be academically correct.
LOM-02	Open and closed electric circuit	Teacher plays animation to show current flow when switch is closed.	Students understand how cell, wire, switch and bulb work in a simple circuit.	Circuit symbols, flow direction and labels must be clear and correct.
LOM-03	Faces, edges and vertices	Teacher shows 3D objects and labels parts during explanation.	Students identify faces, edges and vertices of basic 3D shapes.	Shape labels and visual examples must be mathematically correct.
LOM-04	Conditional decision logic	Teacher runs simple if-else simulation using pass/fail or number comparison.	Students understand true path, false path and output decision.	Logic flow, labels and result must match the condition shown.
LOM-05	Loop repetition	Teacher demonstrates how a loop repeats until a condition is met.	Students understand repeated execution and basic loop output.	Loop steps, condition and output must be technically correct.

6. Learning Objective Quality Criteria

Criteria ID	Quality Criteria
LOC-01	Learning objective must be specific to class, subject and topic.
LOC-02	Learning objective must explain what the Teacher should be able to teach using the content.
LOC-03	Content visuals must support the stated learning objective.
LOC-04	Labels and annotations must be readable on classroom screen, projector or smart board.
LOC-05	Subject Teacher or SME must validate academic accuracy before approval.
LOC-06	Content must be Approved before it appears in the Teacher library.

7. Validation Responsibility Matrix

Validation Area	Responsible Role	Validation Focus
Academic accuracy	Subject Teacher / SME	Correctness of concept, labels and explanation.
Content priority	Principal / Product Owner	Release priority and classroom value.
Visual creation feasibility	3D / Animation / Content Team	Feasibility of animation, 3D view, simulation or visual output.
Platform compatibility	Developer Team	Supported file type, link, browser display and launch behaviour.
Testing readiness	QA Tester	Visibility, launch, status and display validation.
Requirement clarity	Business Analyst	Requirement completeness and traceability to BRD, FRD and Content Requirement Brief.

8. Traceability to Business Need

Business Need	Mapped Learning Objective Support
Teachers need visual support for difficult STEM and Coding concepts.	Each learning objective is linked to visual or simulation-based content.
Teachers need quick classroom-ready content.	Each content item is mapped to class, subject, topic and Teacher usage.
Content must be academically accurate.	Each content item has a validation owner.
Only approved content should be visible to Teachers.	Each content item requires validation and approval before Teacher visibility.
Content Team needs clear input.	Each content item includes objective, visual expectation and validation check.

9. Assumptions

- Initial learning objectives are based on Sprint 0 discovery and Content Requirement Brief.
- Subject Teachers or SMEs will validate academic accuracy before approval.
- Content Team will confirm final animation, simulation or visual format feasibility.
- Product Owner will confirm priority for first release content items.
- Release 1 focuses only on teacher-led classroom display.

10. Dependencies

Dependency ID	Dependency	Owner
DEP-01	Confirmation of first release classes, subjects and topics.	Principal / Product Owner
DEP-02	Availability of Subject Teacher or SME for academic validation.	Principal / Product Owner
DEP-03	Confirmation of final content type and output format.	3D / Animation / Content Team
DEP-04	Confirmation of supported content launch format in web portal.	Developer Team
DEP-05	UAT availability for classroom readability validation.	Teachers / Product Owner

11. Next Step

After this Learning Objective Matrix is reviewed, the Business Analyst will prepare the Content Storyboard / Scene Flow for selected content items and then move towards user stories, acceptance criteria and product backlog preparation.